

Databases

Many people keep jumbled collections of information they’ve gathered, for example a folder of favourite recipes. Computers need a more organized way to store and search data collections saved on them.

What are databases?

Databases are programs that let people store and search data effectively. The most common type of database is the relational database, which stores information in tables. The tables are made up of rows, called records, and columns, called fields. Each table holds details about a particular type of item, so a library database might include a table of books, and a separate table of information on the readers.

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▽ **Keys**
Each record has a field containing a unique value, known as the primary key. This means that records containing the same details, such as two copies of *Matilda*, can be distinguished.

Primary key ID	Title	Author	Out of library	Loaned by
1	<i>Pride and Prejudice</i>	Jane Austen	Yes	Emma Hope
2	<i>Matilda</i>	Roald Dahl	Yes	Surinder Singh
3	<i>Matilda</i>	Roald Dahl	No	Null
4	<i>Frankenstein</i>	Mary Shelley	Yes	James Graham

Searching through databases

Searching for information in a database is called querying. The programming language used for querying relational databases is Standard Query Language, popularly known as SQL. Technically, a database only refers to the collection of organized data stored in a computer. Database management systems such as MySQL, SQL Server, Oracle, and PostgreSQL are the programs that allow users to interact with a database.

▷ **SQL query**
The most common SQL command is SELECT, which retrieves values from the database. Queries can use Boolean operators such as OR and AND to specify which values.

SELECT *
FROM Books
WHERE Author = "Jane Austen"

This symbol means "everything".

SELECT *
FROM Books
WHERE Title = "Matilda" AND Author = "Roald Dahl"

The result for this query would be *Matilda*.

REAL WORLD

NoSQL databases

NoSQL (Not only SQL) is an alternative to the traditional relational databases. It keeps track of items using a variety of methods, including keywords and graphs rather than tables. It is often more suitable for very large data collections, or those stored on the cloud.

